Yaroslav Sergaev

ML Engineer / ML Research Engineer

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ML engineer with research experience in LLMs and multimodal models. Developed a lossless method for faster LLM generation (speculative decoding with a diffusion drafter): on Qwen3-0.6B it gives a 1.44× speedup vs 1.35× for a reproduced baseline. Built an LLM agent with hybrid search and LLM ranking for SberBusiness Online (∼3K WAU), a Whisper-based ASR model that replaced a vendor's solution (∼5M RUB/month saved), and CV models reproduced from papers and compressed for edge devices.

Interests: faster LLM generation, VLM / VLA, RL for LLMs, AI agents.
Target roles: ML Research Engineer / Research Scientist.

Experience

ML Engineer (NLP, LLM), Sber May 2025 - Present

PyTorch | Transformers | LLM | Python | PySpark | FastAPI | Jenkins | OpenShift | Docker

- Built and launched an LLM-based AI agent in the SberBusiness Online web and mobile apps: **∼3K weekly active users**, 9 major releases.
- Designed the agent architecture (<30 s latency SLA): a custom harness of API tools, Schema-Guided Reasoning (SGR), a skills system and a memory sub-agent; new tools and skills plug in via simple YAML/JSON configs.
- Built a two-stage hybrid search for the agent over reference data (regions, industries, niches, retail point addresses): a dedicated candidate-retrieval method per data type (fuzzy + TF-IDF for addresses and regions, dense embeddings + TF-IDF for industries and niches), followed by LLM ranking. **hit@1 0.84 → 0.94, MRR 0.88 → 0.96** on the internal benchmark.
- Built the agent's evaluation system: benchmarks with LLM-as-a-Judge and Agent-as-a-Judge.
- Implemented API contracts and async FastAPI services; deployed via Jenkins and OpenShift.

ML Engineer (DL, NLP, Audio), MTS Exolve (MTT) Nov 2024 - May 2025

ASR (Whisper) | BERT | LLM | RAG | PyTorch | Python | Transformers | Deep Learning | MinIO

- Owned end-to-end training of a Whisper-based ASR model for phone calls with channel noise and multiple speakers: data collection, cleaning and augmentation, full-scale training with QAT (FP8) and inference optimization. The final model (WER/CER ∼10-20%) replaced an external vendor's solution, **saving ∼5M RUB/month**.
- **Raised per-instance LLM inference throughput from 1 to 4-5 RPS** with DeepSpeed, for a target load of 100 RPS.
- Fine-tuned Llama-3-8B with LoRA for a website assistant: SFT on "query → reference answer" pairs and rejection sampling on the model's best generations; added RAG with the Qdrant vector database, which improved the assistant's quality according to expert evaluation.
- Prototyped topic discovery in transcribed dialogues: K-Means clustering of BERT embeddings.

ML Engineer (DL, CV), YADRO Jan 2024 - Nov 2024

Computer Vision | Deep Learning | ONNX | TFLite | PyTorch | Python | Transformers | Bash | Git

- Trained text detection models (TextSnake, DBNet) for edge devices, replacing the original ResNet-50 backbone with MobileNetV3-Small: **∼25× fewer backbone parameters** (23.5M → 0.9M) and over 10× less compute. Separately trained image classification models on MobileNetV3.
- Built a two-tower zero-shot image classifier on MobileCLIP: classes are defined by text embeddings, so new ones are added without retraining the model.
- Reproduced models from research papers, then compressed and converted them to ONNX and TFLite for on-device inference, **retaining 90%+ of original quality** on internal benchmarks.
- Built data labeling pipelines (CVAT, Label Studio).

Projects

FlowDraft 2026

Paper accepted for publication in SMILES School Projects Proceedings (SSPP) 2026 | Skoltech Applied AI Center

Lossless speculative decoding method: a diffusion (flow-map) drafter is embedded in a frozen LLM and trained on its own refinement chain. On Qwen3-0.6B it outperforms a reproduced Orthrus baseline: 2.58 vs 2.20 accepted tokens per cycle and a 1.44× vs 1.35× speedup.

github.com/adelardw/FlowDraft

mdfr-rppgfau 2026

Co-author of a paper that passed Phase 1 review at AAAI 2027

Multimodal deepfake detection: FAU (Swin-T + ME-GraphAU) and rPPG (DeepFakesON-Phys) branches fused via a Q-Former. Found that near-perfect validation and test metrics were inflated: 34% of test clips were augmented copies of training clips, and the model could memorize actors. To make it learn forgery artifacts rather than faces, added MTCNN face cropping, contrastive learning with a memory bank and identity-disjoint splits, grouped by source clip and by FaceNet face clusters. On a held-out set of 53 videos from unseen generators: 0.815 accuracy, 0.844 AUROC.

github.com/adelardw/mdfr-rppgfau

SelfExtensionAgent (SEA) 2026

Pet project

Self-extending, Claude Code-style agent harness on LangGraph: when a task needs a tool it lacks, the agent writes the tool in Python, checks it for safety and correctness (static analysis, LLM review, sandboxed smoke test) and registers it. 70+ tools, long-term memory, MCP, human-in-the-loop approvals; works with any OpenAI-compatible model, including local ones via Ollama.

github.com/adelardw/SelfExtensionAgent

AudioDenoisingNet 2025

MSc thesis, HSE University

Compact speech denoising model: a U-Net (1.85M parameters) on STFT spectrograms with phase correction; processes 8 s of audio in ∼1.2 s on a laptop CPU (RTF ≈ 0.15). Fixed an ISTFT edge artifact without retraining, raising SI-SDR on VoiceBank+DEMAND from 5.2 to 9.5 dB.

github.com/adelardw/AudioDenoisingNet

Education

PhD, Computer Science 2025 - 2028 (expected)

Ivannikov Institute for System Programming (ISP RAS), Moscow

Research topic: multimodal language models.

MSc, Machine Learning & Data Analysis 2023 - 2025

HSE University, St. Petersburg

BSc, Theoretical Physics 2018 - 2023

Lobachevsky State University (UNN), Nizhny Novgorod

SMILES research program 2026

Skoltech Applied AI Center

Summer of Machine Learning at Skoltech; FlowDraft was developed within the program.

Coursework in NLP, CV and ML

Yandex School of Data Analysis (YSDA)

Technical skills

Languages	Python, Bash
Frameworks	PyTorch, PyTorch Lightning, Hugging Face Transformers, PEFT, TRL
Domains	RL, NLP, MLLM (VLM / LLM / VLA), TS, RecSys, AI Agents
LLM & VLM	fine-tuning (LoRA / PEFT), RAG, multimodal fusion (Q-Former)
RL & Alignment	RL, RLHF, DPO, GRPO, RLVR, RLAIF
AI Agents	LangChain, LangGraph, MCP (FastMCP), Agent-as-a-Judge, LLM-as-a-Judge
Search & Ranking	BERT, Qdrant, FAISS, TF-IDF, fuzzy matching, LLM ranking
Model Optimization	quantization, distillation, pruning, speculative decoding vLLM, DeepSpeed, AutoAWQ, TorchAO, ONNX, TFLite, TorchScript/JIT
CV & Audio	OpenCV, torchvision, torchaudio
Data	NumPy, Pandas, PySpark, Redis
MLOps & Backend	Docker, MLflow, Optuna, Git, CI/CD (GitLab, Jenkins), FastAPI

Achievements & languages

Hackathons	Top-3, Gazprom ML Hackathon (digital-twin case)
Languages	Russian (native); English B2: research papers, technical reading and writing